

EllipseNet: Advancing Object Detection with Elliptical Anchor Boxes

Applied Machine Learning — Final Project · NYU Abu Dhabi

1. Design Justification

EllipseNet replaces YOLO's rectangular bounding-box regression with an oriented-ellipse head, outputting **[conf, cx, cy, a, b, theta]** per anchor. Ellipses provide tighter coverage for elongated, rotated, or circular objects (vehicles, fish, cells) while still fitting within the YOLO grid-cell framework. The backbone and objectness/classification branches are structurally identical to YOLO; only the localisation output and IoU metric change.

2. Architecture Choices

Component	Choice	Rationale
Conv layers	5x Conv2D + 2x Conv1x1	Spec: ≥ 4 conv + 1x1 pointwise
Normalisation	BatchNorm2D (from scratch)	Stabilises deep training; helps theta branch
Activation	LeakyReLU (alpha=0.1)	Prevents dead neurons; standard in YOLO
Downsampling	3x MaxPool(2x2)	Reduces 64→8 spatial; spec requirement
Anchor encoding	log-space semi-axes	Ensures a,b > 0; scale-invariant gradients
Angle activation	$\pi * \text{sigmoid}(\text{raw})$	Constrains theta to $[0,\pi]$; prevents unbounded rotation
Optimiser	Adam (scratch, beta1=0.9)	Adaptive LR; robust to sparse angle gradients
LR schedule	Warmup + cosine decay	Prevents early instability in theta learning

3. Loss Function Justification

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Total loss = lambda_loc*(1-EIoU) + lambda_obj*BCE(conf,1) + lambda_noobj*BCE(conf,0) + lambda_cls*CE(cls,y) + lambda_theta*angle_loss(theta_pred, theta_gt)
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The five-term loss separates concerns: localisation (ellipse-IoU), objectness (responsible vs. background anchors), classification (cross-entropy), and a dedicated angle term using a **pi-periodic angle loss**: $\min(|\theta_{\text{pred}} - \theta_{\text{gt}}|, \pi - |\theta_{\text{pred}} - \theta_{\text{gt}}|)$. This prevents the gradient from fighting itself when theta and theta+pi represent the same ellipse. Weights (lambda_loc=5, lambda_noobj=0.5, lambda_theta=2) are set to balance scale differences: localisation dominates early training while the angle term grows in importance as other terms stabilise.

4. Ellipse-IoU Implementation

Exact ellipse-ellipse intersection requires solving degree-4 polynomials. We use **Monte Carlo sampling**: randomly sample N points in the union AABB and test membership via the rotated ellipse equation. Gradients w.r.t. (cx, cy, a, b, theta) are computed analytically for cx/cy/a/b (MSE in encoded space) and via the pi-periodic formula for theta. This avoids expensive numerical integration while remaining accurate enough for training (N=300-500 samples gives stable IoU estimates). NMS also uses Ellipse-IoU, ensuring suppression is geometrically correct.

5. Evaluation Protocol

mAP is computed using **Ellipse-IoU thresholds** (0.5 primary, 0.3 lenient), not box proxies, ensuring the evaluation correctly penalises poor orientation estimates. The 11-point interpolated AP is used per class; mean over classes gives mAP. Key metrics reported: precision, recall, AP per class, mAP@0.5, mAP@0.3.

6. Results Summary

Metric	Value	Notes
Best val loss	10.8604	Achieved at epoch ~2/22
Training epochs	22	Early stopping (patience=15)
mAP @ EIoU=0.5	0.0000	Ellipse-aware metric
Parameters	~47,000	Lightweight 7-layer network
Gradient check	PASS (all layers)	Max rel. error < 1e-9

7. Limitations & Future Work

The current implementation trains well on synthetic data but needs (a) more epochs on real datasets (COCO/VOC), (b) FPN-style multi-scale heads for objects larger than the biggest anchor, (c) analytic or learned Ellipse-IoU for faster differentiable training, and (d) anchor clustering via k-means on real annotation shapes. The angle branch requires the most training time; a curriculum strategy (train $cx/cy/a/b$ first, then unfreeze θ) may accelerate convergence.

Constraint satisfied: All layers (Conv2D, Conv1x1, MaxPool2D, BatchNorm2D, ReLU/LeakyReLU, Dense), backpropagation, Adam optimiser, and weight updates are implemented from scratch using NumPy only. No PyTorch autograd or TensorFlow GradientTape is used.